$$-\left(\cos^2\frac{x}{2}-\sin^2\frac{x}{2}\right)=-\cos\frac{2x}{2}=-\cos x$$

$$\cos 2x+\cos x=0$$

$$2\cos^2 x-1+\cos x=0$$

$$2t^2+t-1=0 \qquad t=\frac{-1\pm\sqrt{1+8}}{4}=$$

$$\frac{-1\pm 3}{4}= \begin{cases}-1\\ \frac{1}{2}\end{cases}$$

$$\cos x=\frac{1}{2} \qquad \cos x=-1$$

$$x=\pm\frac{\pi}{3}+2\pi k \qquad x=\pi+2\pi k$$

$$1\geq\log_5\left(1-x^2-4x\right)\geq 2\cos\left(\frac{\pi}{3}\sin\frac{\pi x}{4}\right)$$

$$1-(x^2+4x+4-4)=1-(x+2)^2+4=$$

$$5-(x+2)^2$$

$$\log_5\left(5-(x+2)^2\right)\leq\log_5 5\leq 1$$

$x = -2$

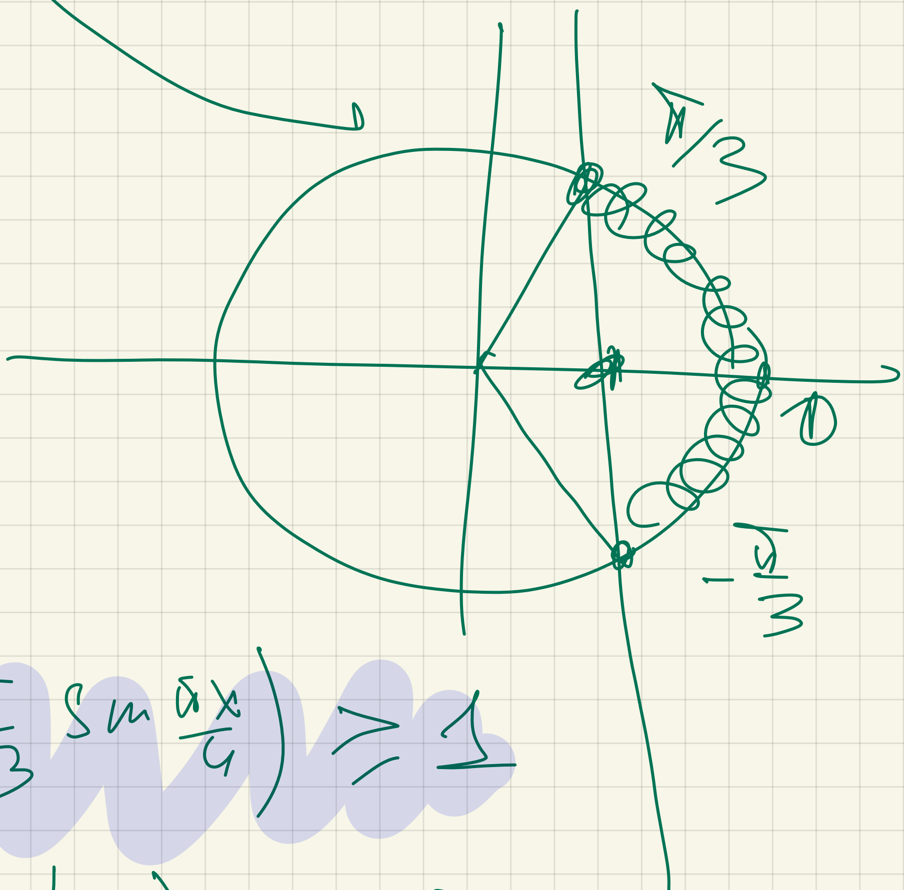
$\varnothing$

$$-1 \le \sin\frac{\pi x}{4} \le 1$$

$$-\frac{\pi}{3} \le \frac{\pi}{3}\sin\frac{\pi x}{4} \le \frac{\pi}{3}$$

$$2\cos\left(\frac{\pi}{3}\sin\frac{\pi x}{4}\right) \ge \frac{1}{2}$$

$\varnothing$



$$\cancel{2\cos\left(\frac{\pi}{3}\sin\frac{\pi x}{4}\right) \ge 1}$$

$$\frac{\pi}{3}\sin\left(-\frac{\pi}{2}\right) = -\frac{\pi}{3}$$

$$\cos\left(-\frac{\pi}{3}\right) = \cos\frac{\pi}{3} = \frac{1}{2}$$

$\frac{1}{2} \cdot 2 = 1$

$X = -2$

$\sqrt{2xy + a} = x + y + 5$

Мет.
рем.

$$\begin{cases} (x+5)^2 + (y+5)^2 = a + 25 \\ x + y + 5 \geq 0 \end{cases}$$

$y \geq -5 - x$

$$\begin{cases} 2^{1+x} = 32a\sqrt{2} \\ \sqrt{x^2 + a^2 + 2 - 2x - 2a} + \sqrt{x^2 + a^2 - 6x + 9} = \sqrt{5} \end{cases}$$

$$\Big\} \ \sqrt{(x-1)^2 + (a-1)^2} + \sqrt{(x-3)^2 + a^2} = \sqrt{5}$$

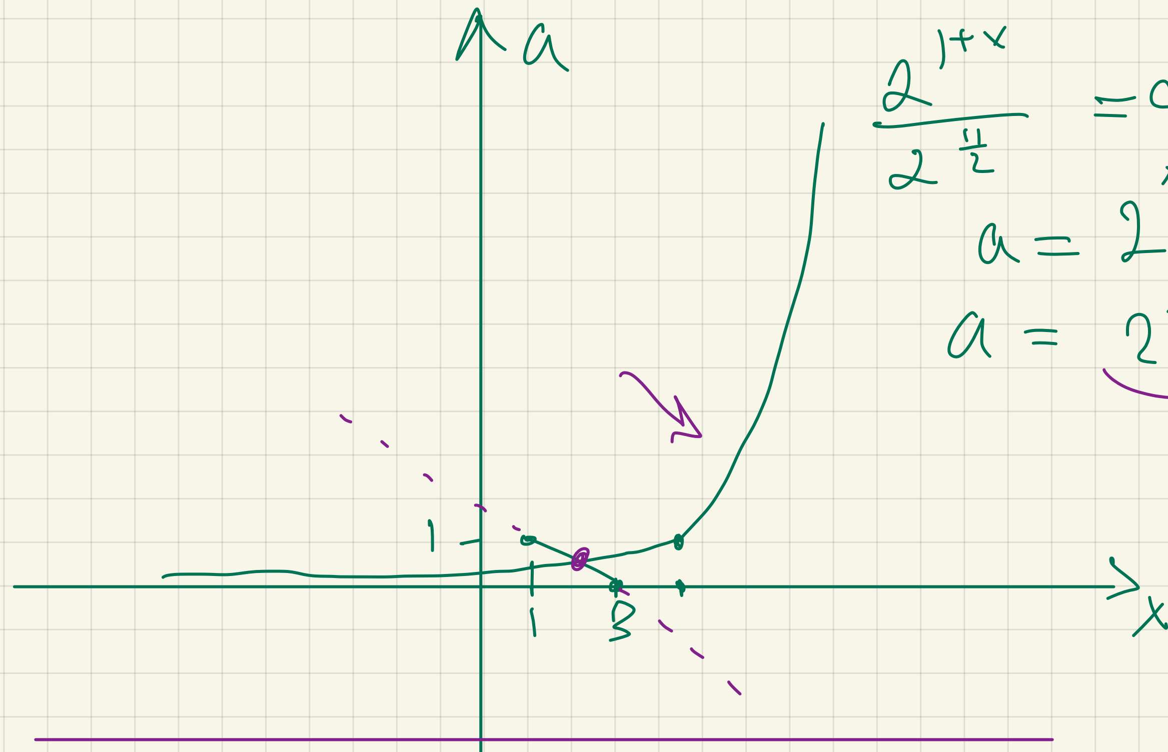
$$2^{1+x} = 32a\sqrt{2}$$

$$2^{1+x} = 2^5 \cdot 2^{\frac{1}{2}} \cdot a$$

$$\frac{2^{1+x}}{2^{\frac{11}{2}}} = a$$

$$a = 2^{x+1-\frac{11}{2}} =$$

$$a = 2^{x-4,5}$$



$y = kx + b$

$0 = 3k + b$

$1 = k + b$

$b = -3k$

$1 = -2k$

$k = -\frac{1}{2}$

$b = \frac{3}{2}$

$$a = -\frac{1}{2}x + \frac{3}{2}$$

$$2^{x-4,5} = -0,5x + 1,5$$

1,5

2

2,5

-3 ?

$\frac{1}{2} \cdot \frac{3}{2}$

$$2^{-3} \overset{?}{=} -0,5 \cdot 1,5 + 1,5$$

$$\frac{1}{8} = -\frac{3}{4} + \frac{3}{2} = \frac{-3+6}{8} = -\frac{3}{8}?$$

$x = 2$

$$2^{-2,5} = -2 \cdot \frac{1}{2} + 3,5$$

$$\frac{1}{\sqrt{2^5}} = -1 + 1,5 = \frac{1}{2}$$

$x = 2,5$

$$2^{-2} = \frac{1}{4} = -2,5 \cdot \frac{1}{2} + 1,5$$

$$-\frac{5}{2} \cdot \frac{1}{2} + \frac{3}{2}$$

$$-\frac{5}{4} + \frac{3}{2}$$

$$\frac{-5+6}{4} = +\frac{1}{4}$$

$x = 2,5$

$a = \frac{1}{4}$

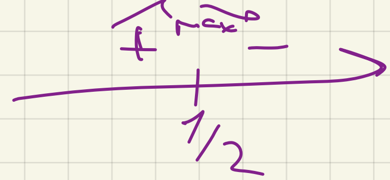
$f(x) = |x - a| - x^2$

наиб зн не меньше 1

1) $f(x) = -x^2 + x - a$ $\quad x \geq a$

2) $f(x) = -x^2 - x + a$ $\quad x < a$

1) $f'(x) = -2x + 1 \rightarrow x = \frac{1}{2}$ max



$$-\frac{1}{4}+\frac{1}{2}-a \geqslant 1$$

$$\frac{1}{4}-a \geqslant 1$$

$$a \leq -\frac{3}{4} \quad \checkmark$$

2) $x_0 = -\frac{1}{2}$ $\qquad -\frac{1}{4}+\frac{1}{2}+a \geqslant 1$

$$a \geqslant 1-\frac{1}{4} = \frac{3}{4}$$

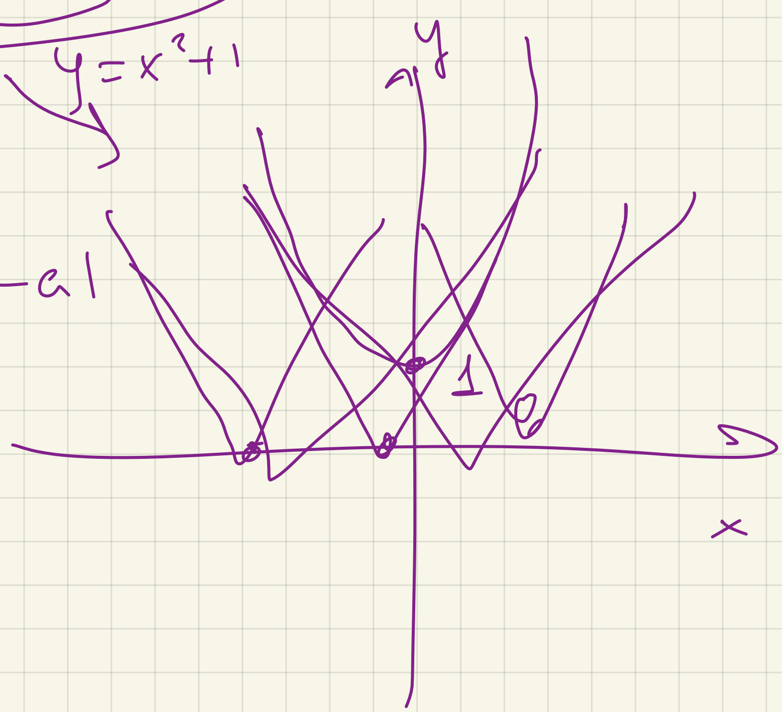
$$a \in \left(-\infty; -\frac{3}{4}\right] \cup \left[\frac{3}{4}; +\infty\right)$$

$$f(x) = |x-a| - x^2 \geqslant 1$$

$$|x-a| \geqslant x^2+1$$

$y=|x-a|$ $\qquad y = x^2+1$

графике $y=|x-a|$



$$x \geqslant a$$

$$x-a \geqslant x^2+1$$

$$a \leq x - x^2 - 1$$

Наим зн больше -24 $D/3$

$$f(x)=4ax+|x^2-6x+5|$$

$$f_1(x)=4ax+x^2-6x+5 \quad x\in(-\infty;1]\cup[5;+\infty)$$

$$f_2(x)=4ax-x^2+6x-5 \quad x\in(1;5)$$

$$f_1'(x)=4a+2x-6=0$$

$$f_2'(x)=4a+2x-6=0 \to x=3-2a$$

$$1<3-2a<5$$

$$2a>-2$$

$$a<1 \quad a>-1$$

1)

$$2a+x-6=0$$

$$1\geq x=6-2a\geq 5$$

$$2a\leq 1$$

$$a\leq \frac{1}{2}$$

$$a\in(-\infty;\frac{1}{2}]\cup[\frac{5}{2};+\infty)$$

$$6-2a\leq 1$$

$$2a\geq 5$$

$$a\geq \frac{5}{2}$$

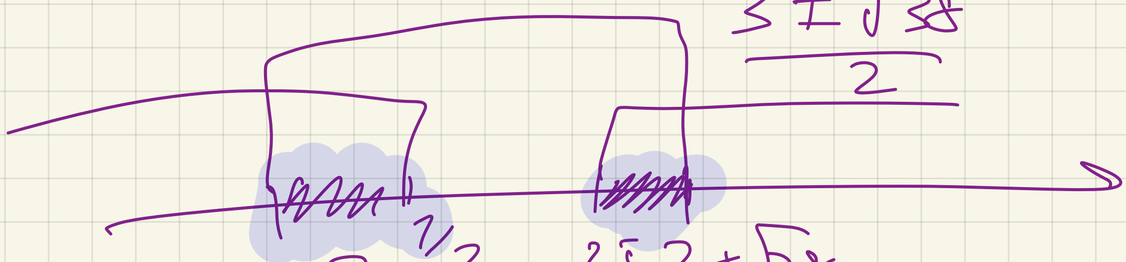
$$4a(6-2a)+(6-2a)^2-6(6-2a)+5>-24$$

$$\cancel{24a}-8a^2+\cancel{36}-\cancel{24a}+4a^2-\cancel{36}+12a+5+24>0$$

$$-4a^2+12a+29>0$$

$$\frac{-12 \pm \sqrt{144+4\cdot 4\cdot 29}}{-8}$$

$$\frac{-12 \pm 4\sqrt{38}}{-8} = \frac{-3 \pm \sqrt{38}}{-2} = \frac{3 \pm \sqrt{38}}{2}$$



$1/2$ $\quad$ $2,5$ $\quad$ $\frac{3+\sqrt{38}}{2} \approx 4,5$

$\frac{3-\sqrt{38}}{2} \approx -1,5$

$$a \in \left[\frac{3-\sqrt{38}}{2} ; \frac{1}{2}\right] \cup \left[4,5 \quad \frac{3+\sqrt{38}}{2}\right]$$

2) $a \in (-1; 1)$